

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Analysis of Training MSE = 0.85 and Testing MSE = 5.20

Introduction

Mean Squared Error, or MSE, is used to measure the difference between the actual student scores and the scores predicted by the neural network.

The formula is:

$$[\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2]$$

Where:

- (y_i) is the actual student score.
- $(\hat{y}_i)$ is the predicted student score.
- (n) is the total number of students.

In this problem, the neural network gives:

[
Training\
]

MSE=0.85
]

[
Testing\
]

MSE=5.20

The training error is very low, but the testing error is much higher. This means that the model performs well on the data used for training, but it does not perform well on new student data.

This situation mainly indicates **overfitting**.

Meaning of the MSE Difference

The difference between the training and testing MSE is:

[
5.20-0.85=4.35
]

This is a large difference.

A training MSE of 0.85 shows that the neural network has learned the training examples very closely. However, a testing MSE of 5.20 shows that the model is unable to generalize properly to unseen data.

The model may have learned not only the important patterns but also the noise and unnecessary details present in the training dataset.

For example, the model may perfectly learn the marks of students in the training set based on attendance, internal marks and study hours. But when details of new students are given, its predictions may be inaccurate.

Possible Reasons for the High Testing Error

1. Overfitting

Overfitting is the most likely reason for the difference.

It occurs when the neural network learns the training data too closely. It may even remember individual training examples instead of learning the general relationship between the input variables and student scores.

For example, a very complex network may memorize the marks of students in the training data. When a new student record is given, the model may fail because it has not learned a general rule.

Effect

- Training error becomes very low.
- Testing error becomes high.
- The model performs poorly on unseen data.

2. Neural Network Is Too Complex

The neural network may contain too many hidden layers or too many neurons.

A large network has a high learning capacity. This can be useful when the dataset is large and complex. However, when the student dataset is small, a very large network may learn unnecessary patterns.

For example, if the dataset contains only 200 student records but the model has many layers and thousands of parameters, the model may easily overfit.

Effect

The model fits the training data accurately but fails to predict new student scores correctly.

3. Insufficient Training Data

The number of student records used for training may be too small.

A neural network requires enough data to understand the general relationship between features such as:

- Study hours
- Attendance percentage
- Previous marks
- Assignment scores
- Internal marks
- Participation
- Family background
- Sleep duration

When the training dataset is small, the model may not see enough types of student behaviour.

Effect

The model may work well only for the limited patterns present in the training set.

4. Training and Testing Data Are Different

The training and testing datasets may have different distributions.

For example:

- The training data may contain mostly high-performing students.
- The testing data may contain many low-performing students.
- The training data may come from one department.

Therefore, even a few incorrect records can greatly increase the testing MSE.

7. Lack of Data Preprocessing

The input features may not have been properly preprocessed.

For example, study hours may range from 1 to 10, while attendance may range from 0 to 100. If these features are directly given to the network without scaling, features with large values may influence the model more.

Missing values, duplicate records and categorical values may also create problems.

Effect

The neural network may learn unstable or incorrect patterns.

8. Too Many Training Epochs

The model may have been trained for too many epochs.

In the beginning, both training and validation errors usually decrease. After a certain number of epochs, the training error may continue to decrease, but the validation error may begin to increase.

This happens because the model starts memorizing the training data.

Effect

The training MSE becomes very small, while the testing MSE increases.

9. No Regularization

Regularization helps control the complexity of a neural network.

If methods such as dropout, L1 regularization or L2 regularization are not used, the model weights may become too large and the model may depend heavily on particular training examples.

Effect

The model becomes sensitive to small changes in input and performs poorly on new data.

10. Irrelevant Input Features

The dataset may contain features that are not useful for predicting student scores.

For example:

- Student ID
- Roll number
- Phone number
- Random code

- Record number

These features do not have a meaningful relationship with academic performance. However, the neural network may still try to find patterns in them.

Effect

The model learns unnecessary information and loses its ability to generalize.

11. Data Leakage

Data leakage occurs when information related to the final student score is accidentally included in the input data.

For example, if the final grade or total result is included as an input feature, the model may obtain very low training error.

However, this feature may not be available when predicting the score of a new student.

Effect

The training result looks excellent, but the model fails during testing or real-world use.

12. Testing Dataset Is Too Small

If the test dataset contains only a few student records, one or two incorrect predictions can produce a large testing MSE.

For example, if only ten students are used for testing and two of them are outliers, the calculated testing error may become very high.

Effect

The testing MSE may not represent the actual performance of the model.

Suitable Corrective Measures

1. Reduce the Complexity of the Neural Network

The number of hidden layers and neurons should be reduced.

For example, instead of using several hidden layers, a simple network can be used:

- Input layer
- One or two hidden layers
- Output layer

A smaller network is less likely to memorize the training data.

Benefit

It encourages the model to learn only the important relationships in the student dataset.

2. Use Dropout

Dropout randomly disables some neurons during training.

For example, a dropout rate of 0.2 means that 20% of the neurons are temporarily ignored during each training step.

This prevents the network from depending too much on a small group of neurons.

Benefit

- Reduces overfitting
- Improves generalization
- Makes the model more robust

3. Apply L1 or L2 Regularization

Regularization adds a penalty to the loss function.

L2 regularization

$$\begin{aligned} [\\ \text{Loss} = \text{MSE} + \lambda \sum w^2 \\] \end{aligned}$$

Where:

- w represents the model weights.
- λ controls the strength of regularization.

L2 regularization prevents the model weights from becoming too large.

Benefit

The model becomes smoother and less sensitive to the training data.

4. Use Early Stopping

A separate validation dataset should be used during training.

Training should be stopped when the validation error stops decreasing.

For example:

- Epoch 10: Validation MSE = 2.1
- Epoch 20: Validation MSE = 1.8
- Epoch 30: Validation MSE = 1.9
- Epoch 40: Validation MSE = 2.3

In this case, training should stop around epoch 20 because the validation error starts increasing after that.

Benefit

Early stopping prevents the model from memorizing the training data.

5. Increase the Training Dataset

More student records should be collected.

The dataset should include students with different:

- Academic levels
- Attendance patterns
- Study habits
- Departments
- Backgrounds
- Performance levels

Benefit

A larger and more diverse dataset helps the neural network learn general patterns.

6. Perform Proper Data Preprocessing

The dataset should be cleaned before training.

The following steps can be used:

- Remove duplicate records.
- Handle missing values.
- Correct invalid values.
- Convert categorical data into numbers.
- Normalize or standardize numerical features.
- Detect and handle outliers.

For standardization:

$$z = \frac{x - \mu}{\sigma}$$

Where:

- (x) is the original value.
- (μ) is the mean.
- (σ) is the standard deviation.

Benefit

All input features are placed on a similar scale, which helps the neural network learn properly.

7. Use a Proper Train-Validation-Test Split

The data can be divided as follows:

[70%\text{for training}]

[15%\text{for validation}]

[15%\text{for testing}]

The records should be shuffled before splitting.

Benefit

- Training data is used to learn.
- Validation data is used to tune the model.
- Testing data is used only for final evaluation.

8. Use Cross-Validation

K-fold cross-validation can be used when the dataset is small.

In 5-fold cross-validation, the dataset is divided into five equal parts. Four parts are used for training and one part is used for validation. This process is repeated five times.

Benefit

It gives a more reliable estimate of the model's performance and reduces the effect of an unlucky data split.

9. Remove Irrelevant Features

Unnecessary features such as student ID, roll number or phone number should be removed.

Useful features may include:

- Attendance
- Study hours
- Previous marks

- Assignment scores
- Internal marks
- Class participation

Feature selection can be performed based on correlation, domain knowledge or model importance.

Benefit

The model becomes simpler and focuses on meaningful information.

10. Handle Outliers Carefully

Outliers should be detected using methods such as:

- Box plot
- Z-score
- Interquartile range
- Manual inspection

An outlier should not be removed immediately. First, it should be checked to determine whether it is a genuine student record or a data-entry mistake.

Benefit

Correct handling of outliers can reduce the testing MSE and improve prediction stability.

11. Tune Hyperparameters

The following hyperparameters should be adjusted carefully:

- Learning rate
- Number of epochs
- Batch size
- Number of hidden layers
- Number of neurons
- Dropout rate
- Regularization strength

A very high learning rate may cause unstable training, while a very low learning rate may make the training process slow.

Benefit

Proper hyperparameter tuning helps the model achieve a balance between learning and generalization.

12. Check for Data Leakage

Input variables should be checked carefully.

Any feature that directly or indirectly reveals the final student score should be removed.

For example, the following should not be used as input:

- Final grade
- Result status
- Final total
- Pass or fail label

Benefit

It ensures that the performance of the model is genuine and realistic.

Expected Improvement

After applying the corrective measures, the training MSE may increase slightly, but the testing MSE should decrease.

For example:

[Training\MSE=1.20]

[Testing\MSE=1.60]

This result is better than:

[Training\MSE=0.85]

[Testing\ MSE=5.20]

Although the new training error is slightly higher, the difference between training and testing errors is much smaller. This means the model can perform better on new student data.

Final Analysis

The neural network has a very low training MSE of 0.85 and a high testing MSE of 5.20. The difference of 4.35 clearly shows that the model does not generalize well.

The main reason is likely overfitting. Other possible causes include a complex model, limited training data, noisy records, poor preprocessing, irrelevant features, improper data splitting and the absence of regularization.

The problem can be corrected by reducing the network size, applying dropout and L2 regularization, using early stopping, collecting more data, performing proper preprocessing, using cross-validation and selecting only useful features.

The aim should not be to obtain the lowest possible training error. The actual aim is to create a model

that gives accurate predictions for new and unseen students.

Therefore, a small and balanced difference between training MSE and testing MSE is more important than achieving an extremely low training MSE.

Code of Conduct:

I M.Ajitha(192525391)____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in black ink, appearing to read 'Ajitha', is shown within a rectangular frame.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided